

Successful Use of RFID for Asset Management and SCM

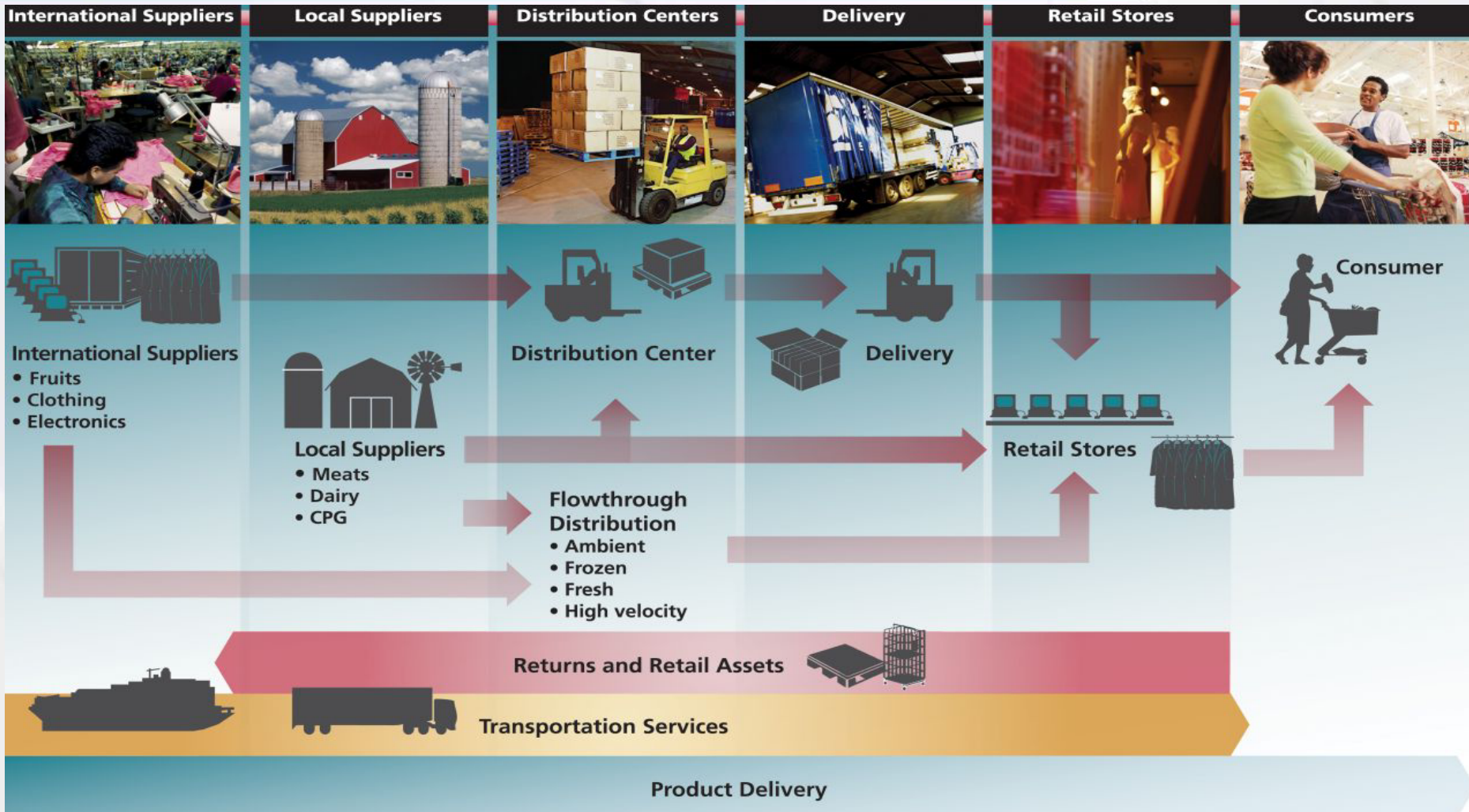


Goh Eng Koon
Senior Vice President
Savi Technology Asia



Asset Management In Supply Chains

Savi Expertise



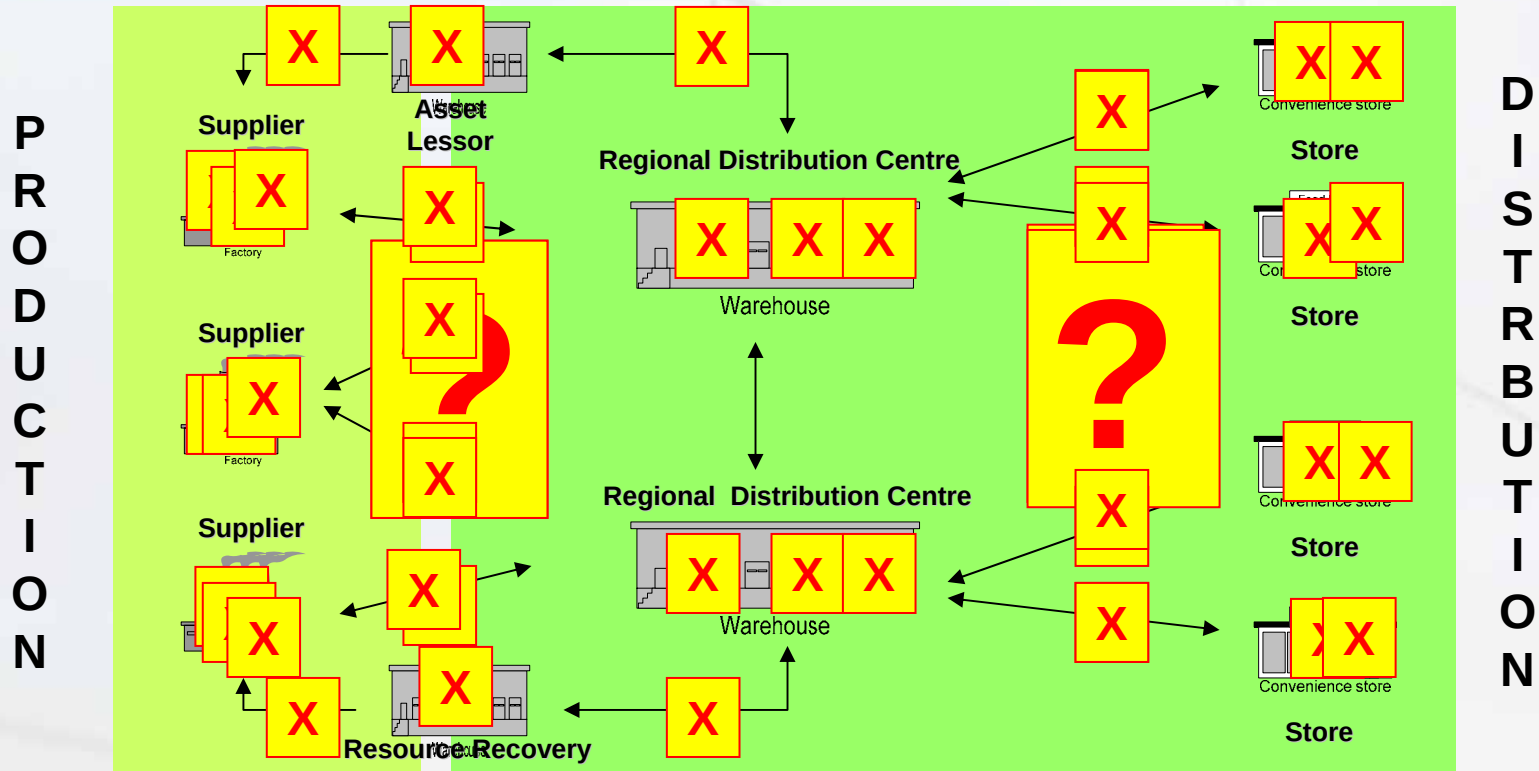
Visibility & Management of Diverse *Transportation Assets* & Their Contents across the Supply Chain Is Inherently Problematic Resulting in High Operating Cost

Business Environment & Pressures for Focusing on Transportation Asset Management

- Increasing competitive pressure to remove cost inefficiencies and to improve profitability
- Governmental legislation to reduce packaging waste drives increased adoption of expensive reusable transportation assets
- Continued sophistication and automation of logistics and distribution activities
- Industry initiatives to improve food traceability from 'farm to fork'
- 'Customer' specific issues & challenges
 - Asset pooling initiatives; value added service offering



Asset Management Pain Points



- Insufficient assets to meet distribution demands
- Hoarding or inadvertent stockpiling of assets at supply points
- Loss of assets in supply chain
- Shrinkage, spoilage and asset damage
- Poor food chain traceability
- Order planning and store delivery reconciliation issues
- High expedite and repositioning transportation costs
- Emergency asset purchase, rental, lease

How can this situation be improved ?

- Improving Visibility & Traceability through the Supply Chain
 - Introduce AIDC at the asset level and at key SC transfer points
- Improving Forecasting & Planning of Asset Requirements
 - Deploy planning tools to utilize accurate AIDC captured data
- Improving Asset Inventory Utilization
 - Reduce asset fleet by using accurate visibility data to improve planning and control
- Increasing Distribution Process Automation
 - Speed up processes with automatic data capture and verification
- Reducing Unnecessary Manual Labour Intervention
 - Use newly acquired AIDC data to reconcile asset & product movements

What an AIDC powered Transportation Asset Mgmt. System Can Provide:

- Provide accurate and timely data for fact based action :
 - Identify which handling locations & parties cause most asset damage
 - Identify parties and locations where most product spoilage occurs
 - Identify which parties have custody where most shrinkage happens
 - Identify which handling locations hoard or inadvertently stockpile assets
 - Identify where assets are lost and never returned
 - Determine optimum asset fleet size to meet actual demand
 - Highlight locations of slow processes and excessive transit or dwell times
 - Highlight and reconcile order and delivery discrepancies
 - etc...

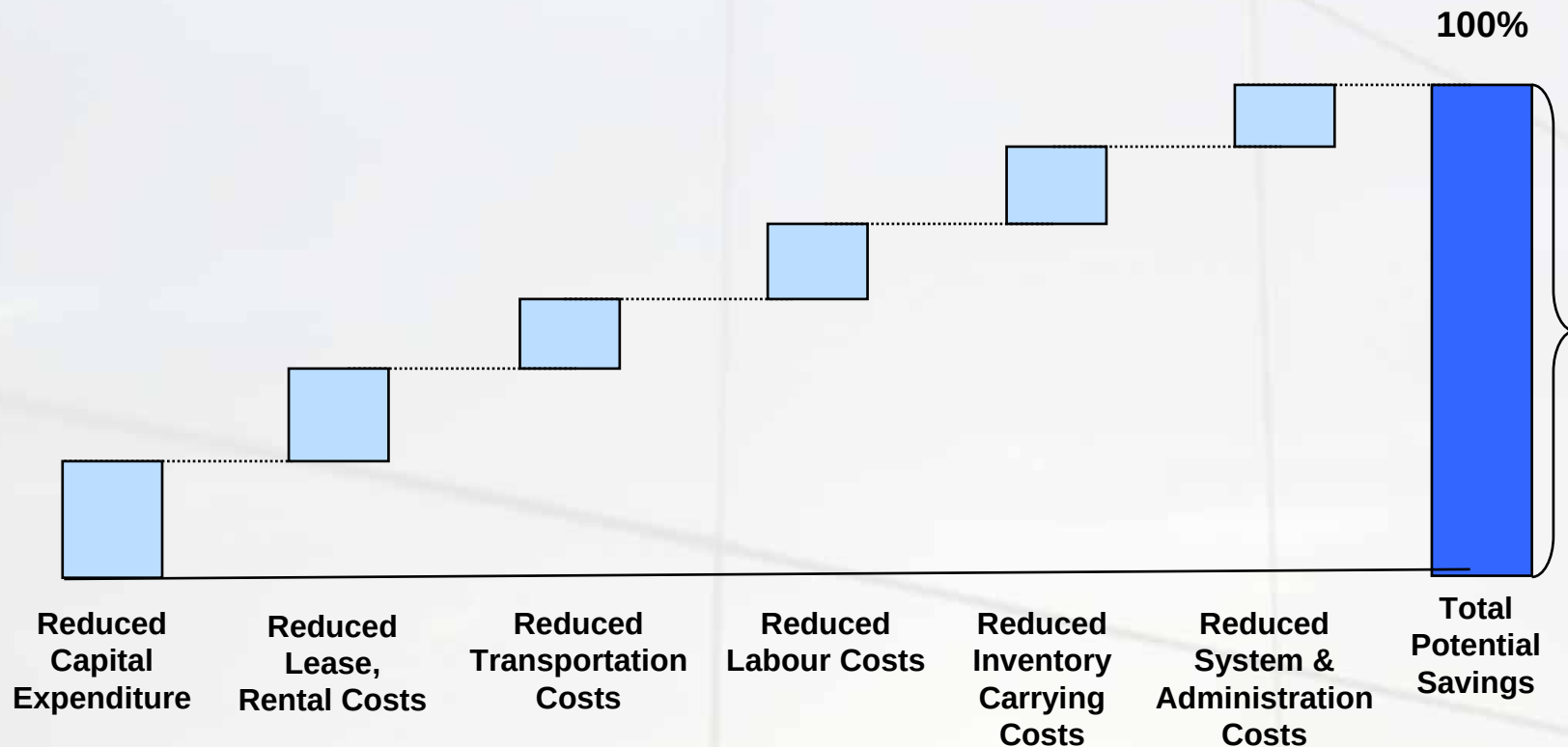
Financial Benefits of Improved Transportation Asset Management

- **Increased Capital Effectiveness**
 - ***Reduced Asset Inventory for Asset Owners***
 - Improved Inventory Utilization, Visibility, and Forecasting
- **Increased Operating Margin**
 - ***Reduced Lease, Rental & Demurrage Charges for Asset Lessees***
 - Improved Inventory Utilization, Visibility, and Forecasting
 - ***Reduced Expediting Transportation Costs***
 - Improved Visibility and Forecasting
 - ***Reduced Shrinkage, Spoilage, Asset Damage & Loss***
 - Improved Visibility
 - ***Reduced Inventory Carrying Costs***
 - Smaller Asset Fleet
 - ***Reduced Labor Costs***
 - Increased Process Automation & Reduced Manual Intervention
- **Opportunities to Increase Revenue**
 - ***Reduced Cost Base***
 - Smaller Asset Fleet
 - ***Improved Merchandise Availability to Stores***
 - Increased Process Automation & Reduced Manual Intervention

Key Cost Benefit Areas of Improved AIDC* Enabled Asset Management

Percentage savings and improvement will vary from customer to customer dependant on key areas of improvement, value & volume of assets deployed.

You should certainly assess the Return on Investment potential



Savi Introduction



Proven RFID Solutions Driving Value

Market Leadership Position

- Fifteen Years Experience With RFID & AIDC Supply Chain Solutions
- Proven Fifth Generation RFID Site, Enterprise And Application Software
- Proven Sixth Generation Standards-based Active RFID Hardware
- Nine Years Experience Passive RFID
- 13 patents issued, 25 more in process

World's Largest RFID Supply Chain Implementations

- Defense Supply Chains
 - Preferred Provider Of Active RFID To The US DoD (1994),
 - Interoperable Global Defense Network – US DoD, UK Mod, NATO, Denmark
 - 35,000 Containers Tracked Daily, 1500+ Sites In 46 Countries
- Closed Loop RFID Based Networks
 - Supply Chain Asset Management
- Ocean Cargo Visibility And Security



- Joint Venture with Hutchison Port Holdings
- Deploying Global Ocean Cargo RFID Network
- Transaction Based Visibility and Security Information Services



What We Do

Proven RFID Solutions Driving Value

- Supply Chain Application Software
 - AMS – Logistics Asset Management
 - TSS – Transportation Security
 - CMS – Consignment Management System
- Real-time Enterprise RFID Platform
 - SmartChain Enterprise Platform
- Active RFID Technology
 - EchoPoint Long Range Active RFID
- Implementation Services
 - AIDC Hardware & Software Integration & Implementation



Corporate Profile

- History
 - Founded In 1989, Acquired By Texas Instruments (November 1995), Sold To Raytheon (1996-97), Management Buy-out (1999)
- RFID Solutions Focused On Defense Supply Chain, Asset Management And Ocean Cargo Visibility And Security
- 230 Employees Worldwide. Head Office In Sunnyvale, CA
- Regional Offices: Alexandria, Va; Singapore; London; Melbourne
- Experienced Management Team
- Two Areas Of Business



Enterprise Solutions For Asset Management
And Defense Markets

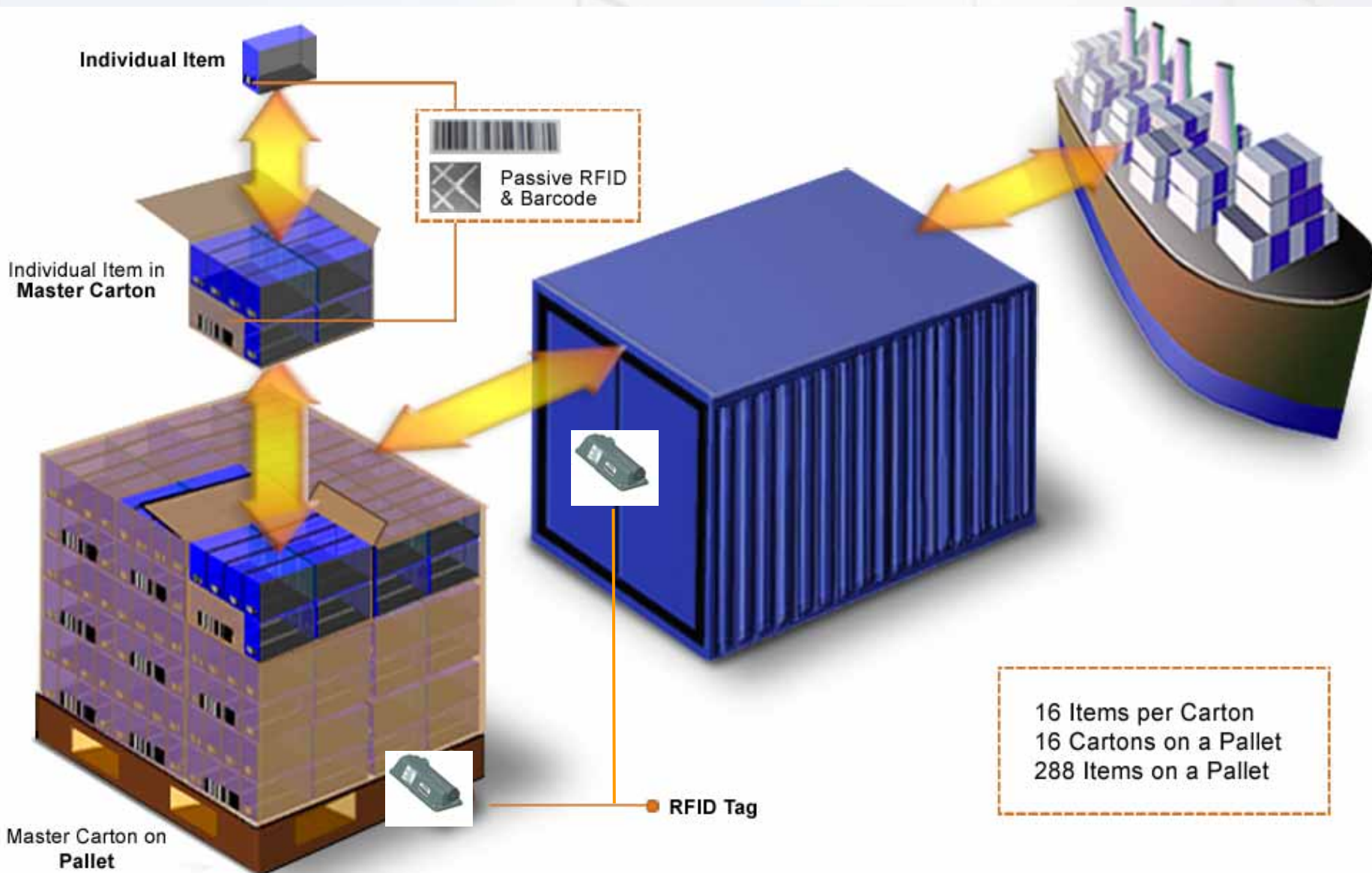


Joint Venture With Hutchison Port Holdings
Deploying Ocean Cargo Visibility And Security Network

Savi History

- 1989 - Begins Operations
- 1991 - Wins DARPA Grant For Applied Research And Development For RFID
- 1992 - First Introduces “Smart Container” Technology
- 1994 - Awarded First US DoD Contract For RFID
- 1995 - Begins Implementation of DoD “Total Asset Visibility” Network
 - Acquired By Texas Instruments
- 1997 - Wins Second RFID Contract With US DoD
- 1999 - Management Buys Company From Raytheon
 - Completes First Commercial Implementation
- 2000 - “SmartChain” – First Commercially Available RFID Software
- 2001 - EchoPoint – A Fifth Generation Active RFID Technology Is Introduced
- 2002 - System Integrator On First Operational Test Of EPC Technology With AutoID
 - Savi And 14 Partners Launch Smart And Secure Tradelanes
- 2003 - Wins Third RFID Contract With The US DoD
- 2004 - Ships 1,000,000th ST-410 Tag To The US DoD
 - SmartChain Rated Best RFID Middleware By Forrester
 - ABI Names Savi A “Global Leader In RFID”
- 2005 - Former Secretary Homeland Defense Tom Ridge Joins Board
 - Savi Networks Joint Venture With Hutchison Port Holdings

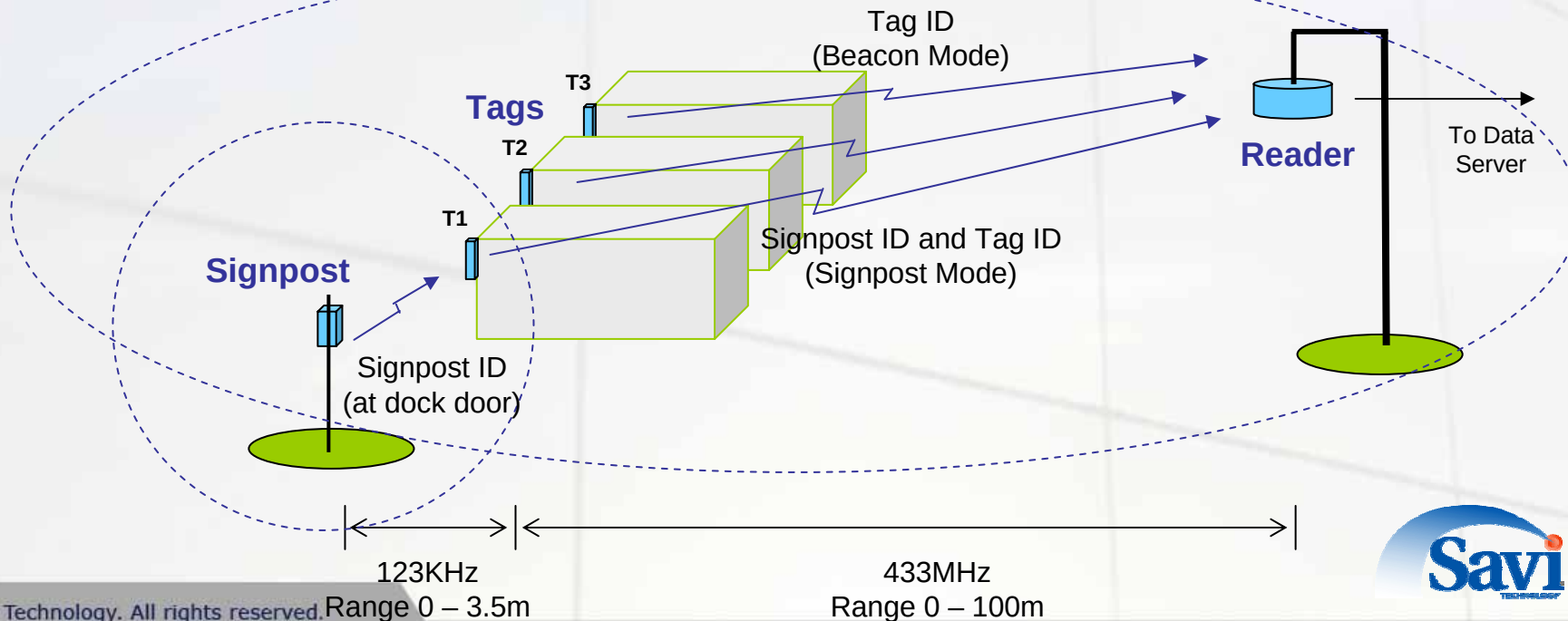
Nested Visibility



EchoPoint™

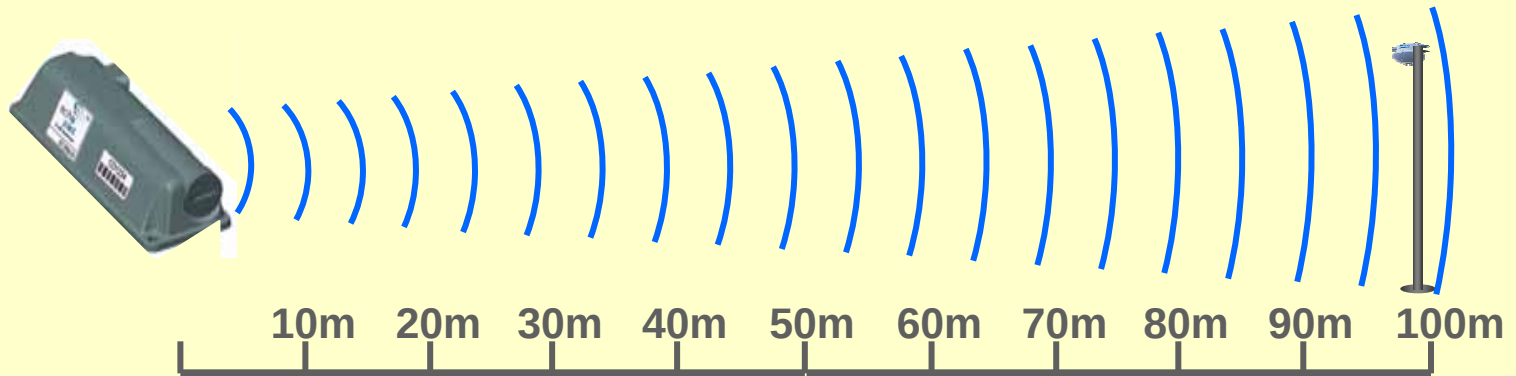
RFID system overview

- EchoPoint tags use two types of RFID technology:
 - Low frequency, short-range RFID technology to determine specific location information (e.g. dock-door or other chokepoint)
 - High frequency, long-range RFID technology to transmit that data over longer distances
- Low frequency (short-range) mode
 - Signpost transmits its signpost ID to the tags in signpost's (i.e. chokepoint) field
 - Signpost transmission distance can be varied
- High frequency, long-range (beacon) mode
 - Tags periodically transmit their tag IDs to the reader (even in the absence of a signpost)
 - Tags transmit their tag IDs and the signpost ID to the reader
 - Tags' beacon can be turned On or Off and beacon rate can be adjusted



Savi's EchoPoint ® Active RFID Technology

Battery Enables Savi Active Tags To Transmit Up to 100M To Readers



Signpost Triggers Tag Transmission Identifying Spot Location



Tag Can Travel Outside Signpost Range While Transmitting to Reader

EchoPoint Product Suite

Dual-Frequency Active RF Tags



EchoPoint Tag 602



**EchoPoint 604/640
Container Tag**



**EchoPoint Security Tag
645 SealTags**

Readers



Fixed reader
Reads all Savi tags @ 100m



HandHeld Mobile Readers

“Signposts” for spot-level location of assets



**Fixed signpost – e.g., to
demarcate dock door
chokepoints**



**Mobile signpost – e.g., to
mark temporary locations
of interest**

Extended Family of RFID Technologies

Savi Tag ST-676
Cargo Container
Security Tag



E.J. Brooks
E-Seal



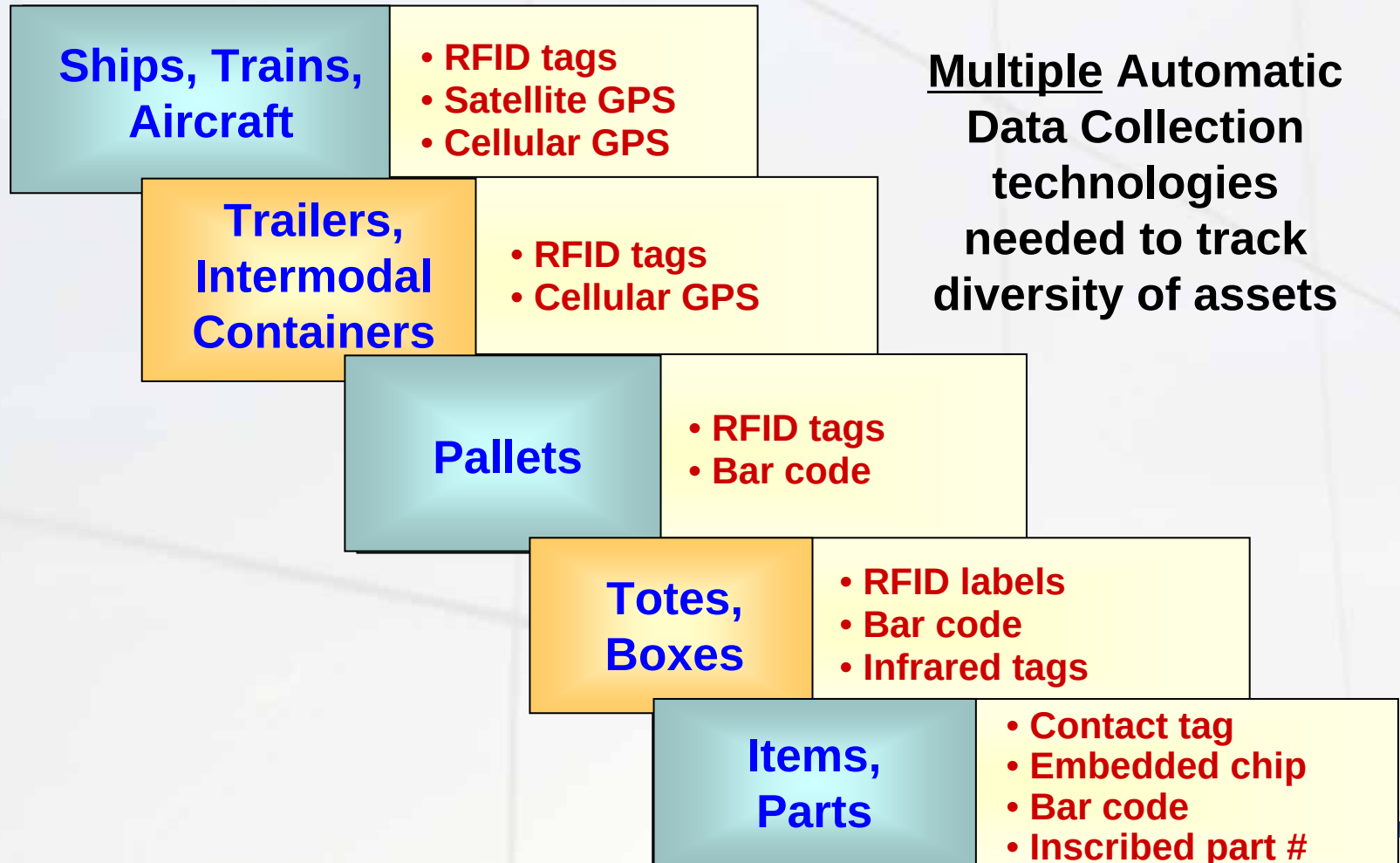
Intermec
Handheld
Reader



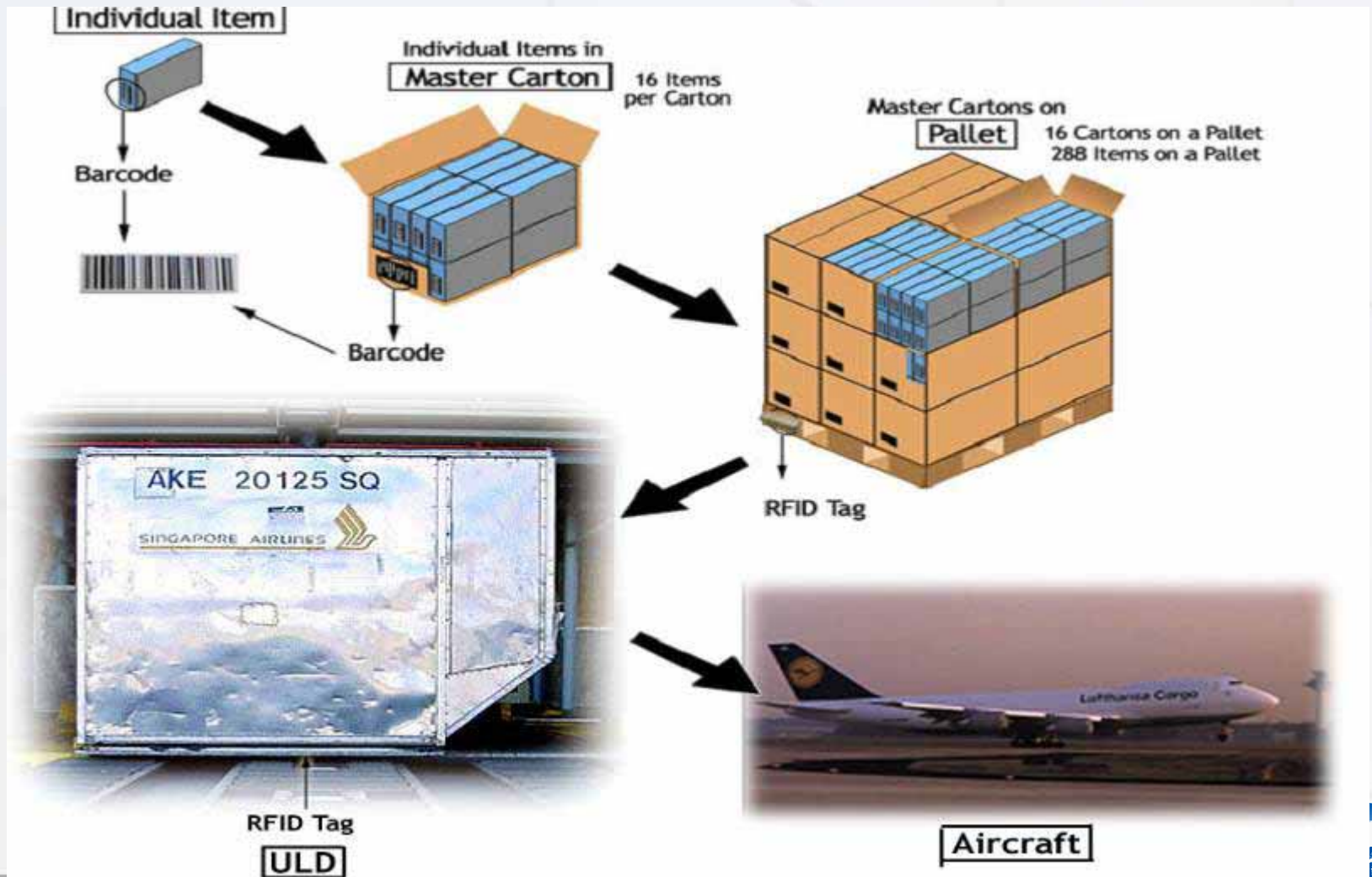
Savi Mobile
Reader
SMR-650
(Symbol III)



Tags for different requirements

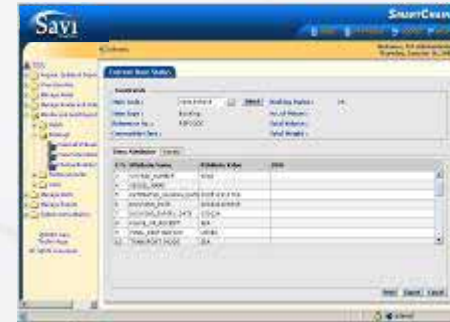


Tracking by Association

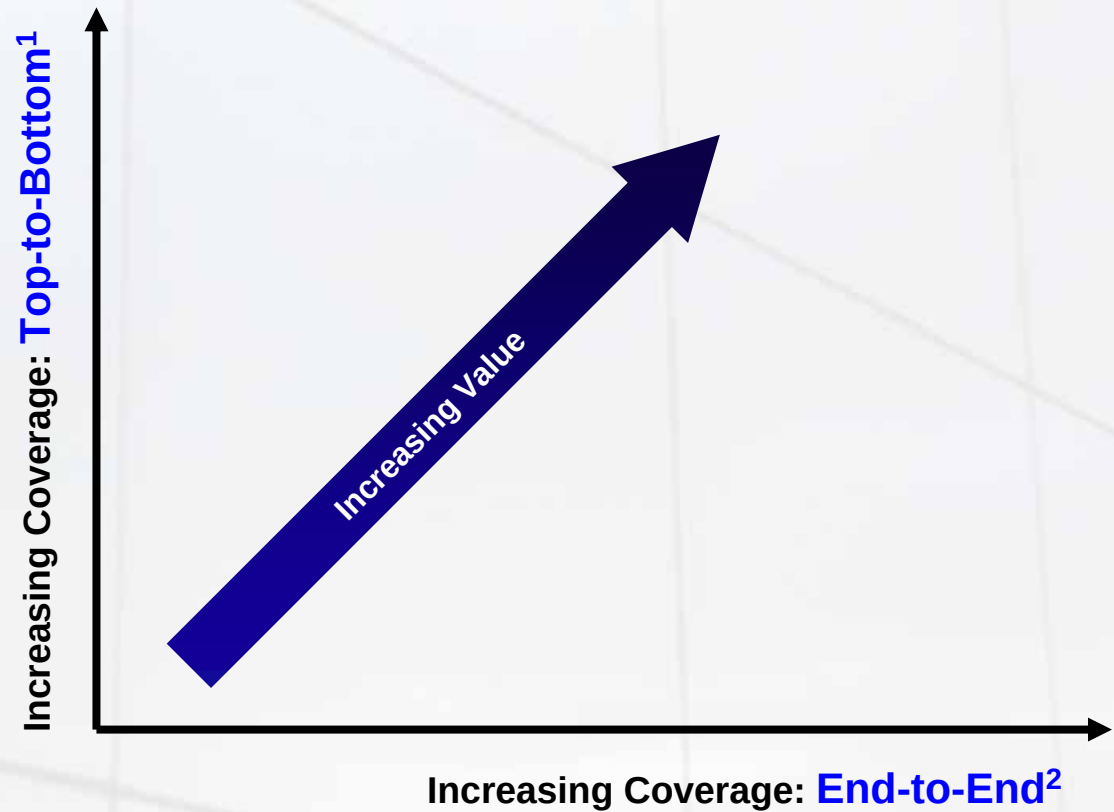
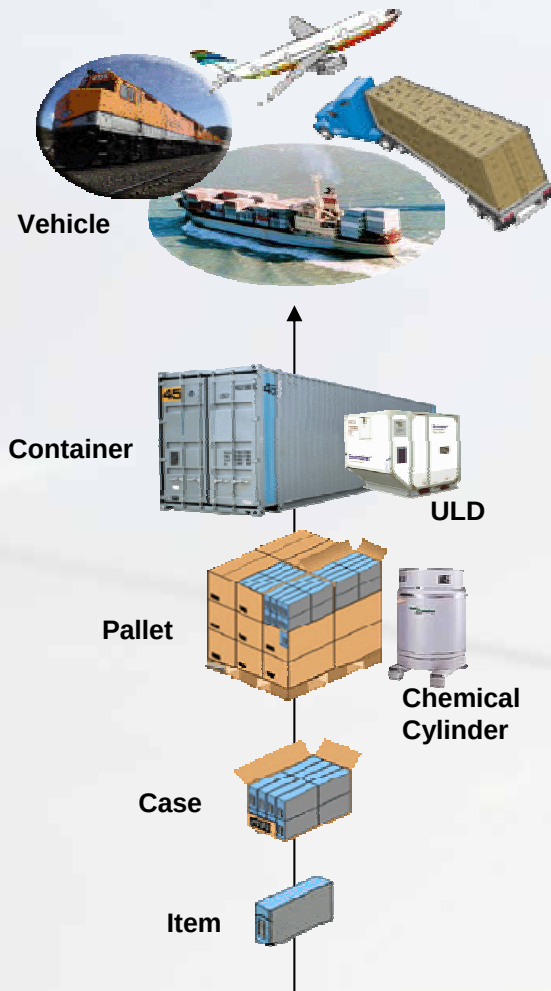


Nested Visibility

Integrated, Global, Multi-Modal



Value is Proportional to Depth¹ & Breadth² of T-VAC



AIDC Solutions Architecture

Applications:

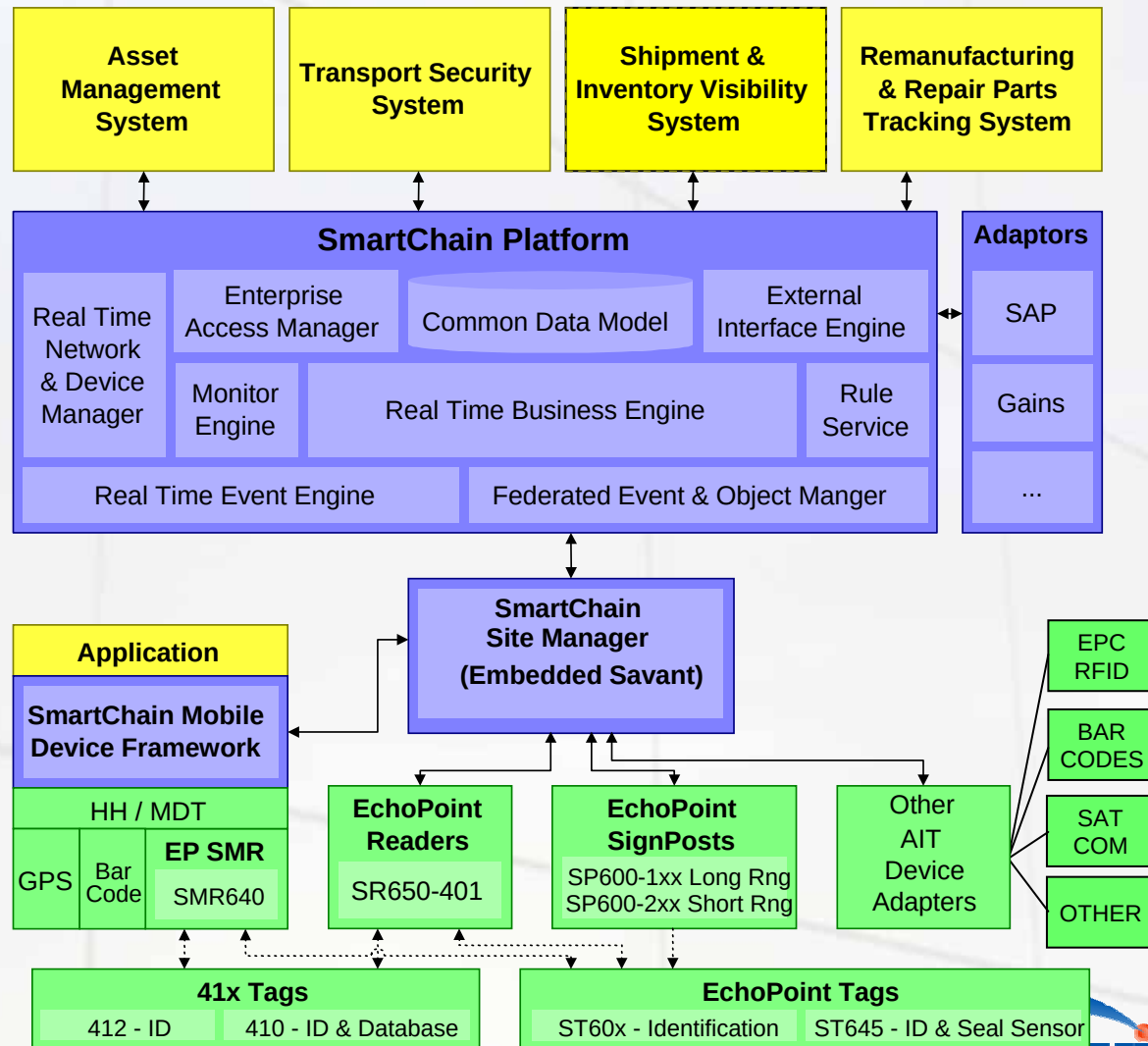
- Solving business problems for organizations dealing with high value or mobile items

System Platform:

- Common Data Model
- Event Management
- Network Management
- System Integration

Network:

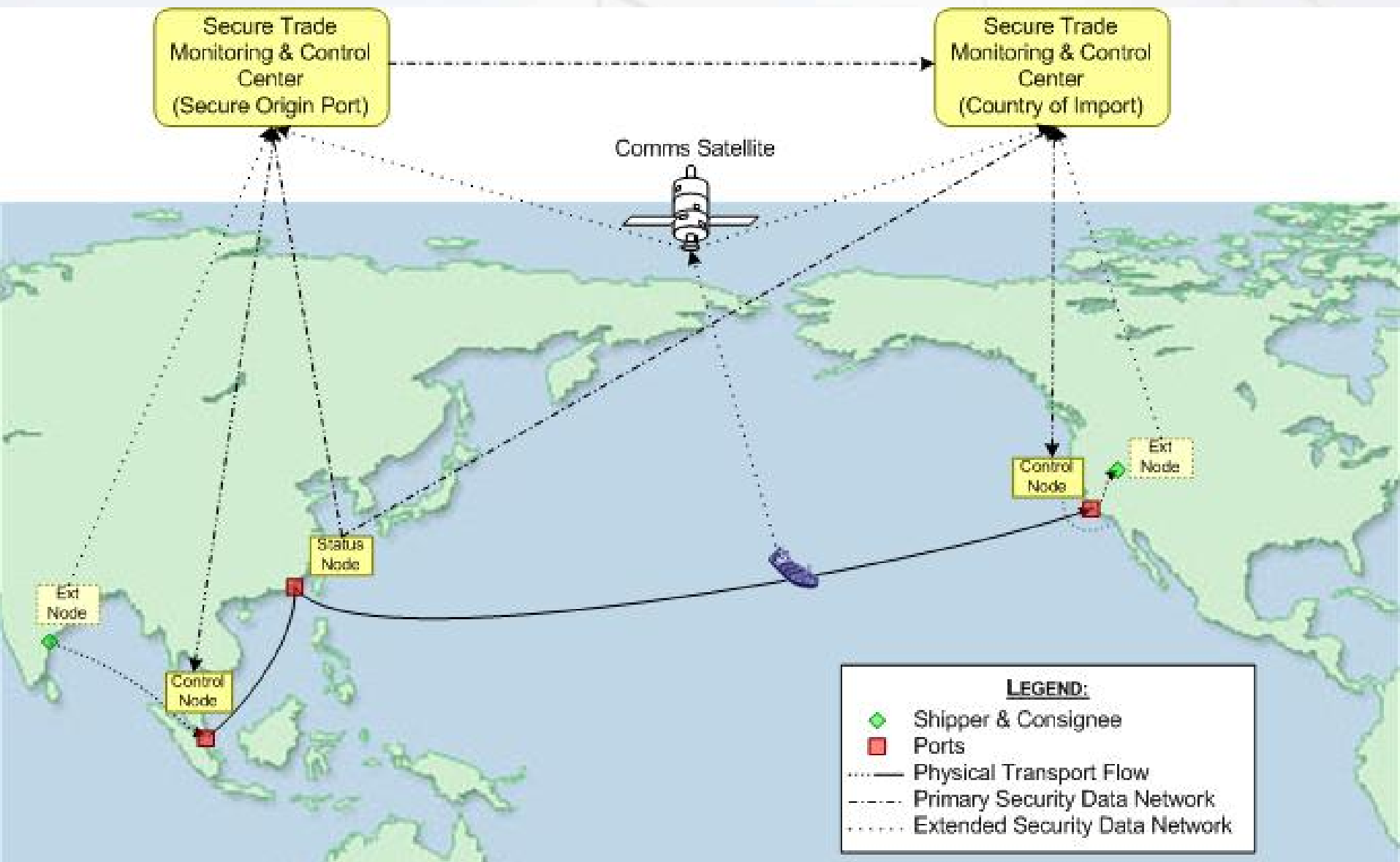
- Distributed Data Management
- Local Transactions
- Real Time Data Capture
- Process Control
- Tags & Sensors



Some Case Studies



Smart & Secure Tradelanes



Solution Baseline: An end to end network for tracking and security



1. Manufacturer/Consolidator

- Associate manifest with container and seal ID
- Verify user and electronically secure container
- Automatically record departure (optional)
- Transmit data to TSS



2. Port of Loading - Arrival

- Automatically record container arrival
- Automatically verify security status at entry gate

3. Port of Loading - Vessel load

- Verify clear for loading status
- Automatically verify security status at quay cranes



Transportation Security System Platform



- Transmit documentation and updates to Customs
- Provide alerts if container deviates from plan
- Provide reports and other analytics for supply chain and security inefficiencies
- Maintain audit trail of accountability
- Provide web access to users



4. Vessel Departure



5. Port of Discharge - Vessel Unload

- Verify clear for unloading status
- Automatically verify security status at quay cranes



6. Port of Discharge - Departure

- Automatically record departure
- Automatically verify security status at exit gate



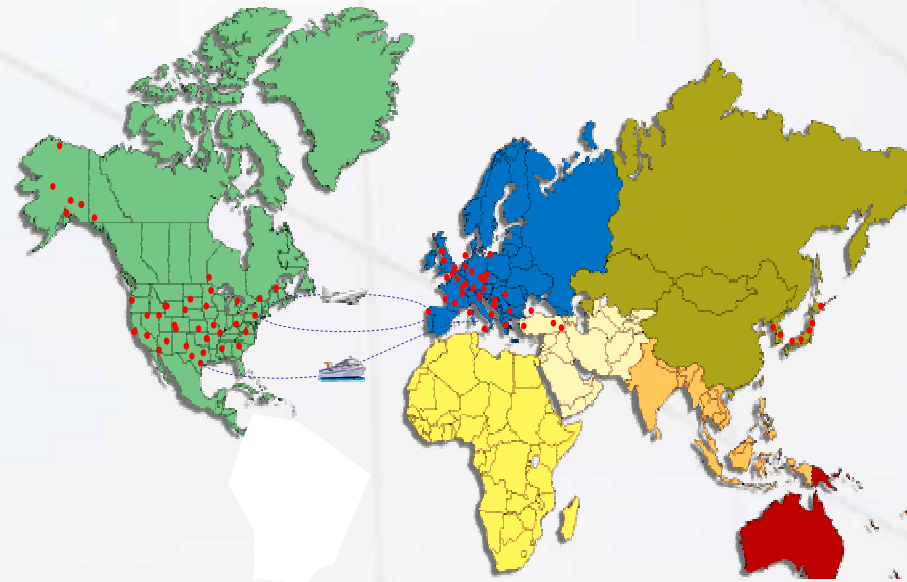
7. Deconsolidation center

- Automatically record arrival (optional)
- Automatically verify security status
- Verify user and unseal container
- Transmit data to TSS

US Department of Defense

Total Asset Visibility Network

- Business Problem: Poor visibility of containers/supplies to theatre. Lost manifests, searching for items for days.
- Solution: Global visibility network with RFID solutions for tracking materials. Integrated with complementary technologies (sensors, GPS, scanners, biometric, and handheld devices).
- Project Brief: Open network, international (ISO) and technology standards-based system.
- Lessons Learned: Total Asset Visibility. Reduction in containers shipped for recent gulf war. Enabled automation of inventory management, transportation, and maintenance capabilities – resulting in reduced cycle time, improved accountability, enhanced security, and improved utilization of resources



- **45+ Countries**
- **800+ Nodes** - airports, seaports, rail terminals consolidation points
- **37,000+ Conveyances/Day**





AFSILC Process Overview

STEP 5: LIFT TO VESSEL

- Record container # and stow location in AFSILC
- Generate Stow Location Report
- Send stow location data from AFSILC to WPS via ftp to complete the Ocean Cargo Manifest.





USF
007-64
226

100-101-01
25-10-000

100-101-01
25-10-000

USF
007-61
225

100-101-01
25-10-000

100-101-01
25-10-000

100-101-01
25-10-000

100-101-01
25-10-000

100-101-01
25-10-000

100-101-01
25-10-000

USF
012147
2269

PROPERTY OF
THE U.S. GOVERNMENT

EMPTY LIFT
ONLY

EMPTY LIFT
ONLY

USF
012173
2269

PROPERTY OF
THE U.S. GOVERNMENT

EMPTY LIFT
ONLY

EMPTY LIFT
ONLY

USF
012201
2269

PROPERTY OF
THE U.S. GOVERNMENT

USF
012229
2269

PROPERTY OF
THE U.S. GOVERNMENT

Corpus Christie Army Depot

Remanufacture & Repair of Military Aircraft

- Business Problem: Annual loss of parts/equipment is high. High labor costs searching for parts.
- Solution: Asset Management Solution and RFID Network provide inventory and item level tracking of aircraft parts. Wireless Network of Readers record part arrival and departure at each shop.
- Project Brief: Install facility-wide item location tracking. Provide location tracking through the repair cycle process. Integrated tracking of components and end items. Enhance repair cycle management and efficiency – maintain database of repair history, cost of repair, and configuration.
- Lessons Learned: Reduced annual investment in locating parts by 26 man years per year.



- **250,000 Parts**
- **151 Shops**
- **3500 Support Equipment**
- **7 Square Miles**

Woolworths

UK Retailer - Closed Loop Distribution

- Business Problem: High volume of theft. 10-15% shrinkage in some product categories.
- Solution: Asset Management solution tracks dolly's movement through DC to correct store, and GPS tracks vehicle on route. Driver uses handheld to confirm delivery.
- Project Brief: UK Home Office sponsored initiative for theft prevention- included the integration of Bar Codes, Active RFID and GPS systems into a single supply chain visibility solution.
- Lessons Learned:
 - Successful tracking of goods between DC and stores
 - Reduction in theft and loss
 - Elimination of incorrect deliveries of dollies to stores
 - Improved distribution minimized costs, reduced stock-outs, and improved top-line revenue goods
- 2003 European Retail Systems Award



- **16,000 Dollies**
- **90 Stores in the UK**



Refrigerated ULDs

- Business Problem: Lack of visibility into location and status of ULD's required for growth.
- Solution: Asset Management System enables Envirotainer to electronically pool and reassign its ULDs from multiple airlines and locations.
- Project Brief: Open network, international (ISO) and technology standards-based system.
- Lessons Learned: Enhanced operational efficiency, asset utilization and customer service.

- **3,000 Temperature Controlled ULD's**
- **Over 60 International Airports**
- **Over 35 Countries**



Envirotainer[®]



South African Railroad

- Business Problem: Unable to provide wagons to customer for movement of materials. Poor visibility of assets.
- Solution: Asset Management Solution and RFID Network provide data on status and location of rolling stock.
- Project Brief: Pilot phase for specific cargo and tradelanes, moving to roll-out phase. Partnership with local companies and support from UN Conference on Trade and Development (UNCTAD).
- Lessons Learned: The system improves asset utilization, customer service, analysis, security, and on-time deliveries.



- **75,000 Railcars**
- **3600 Locomotives**
- **19,000 Miles of track**



Savi SmartChain ® Asset Management Solution for the Retail Industry

Solution Overview



What components are required to build a Supply Chain Asset Management solution?

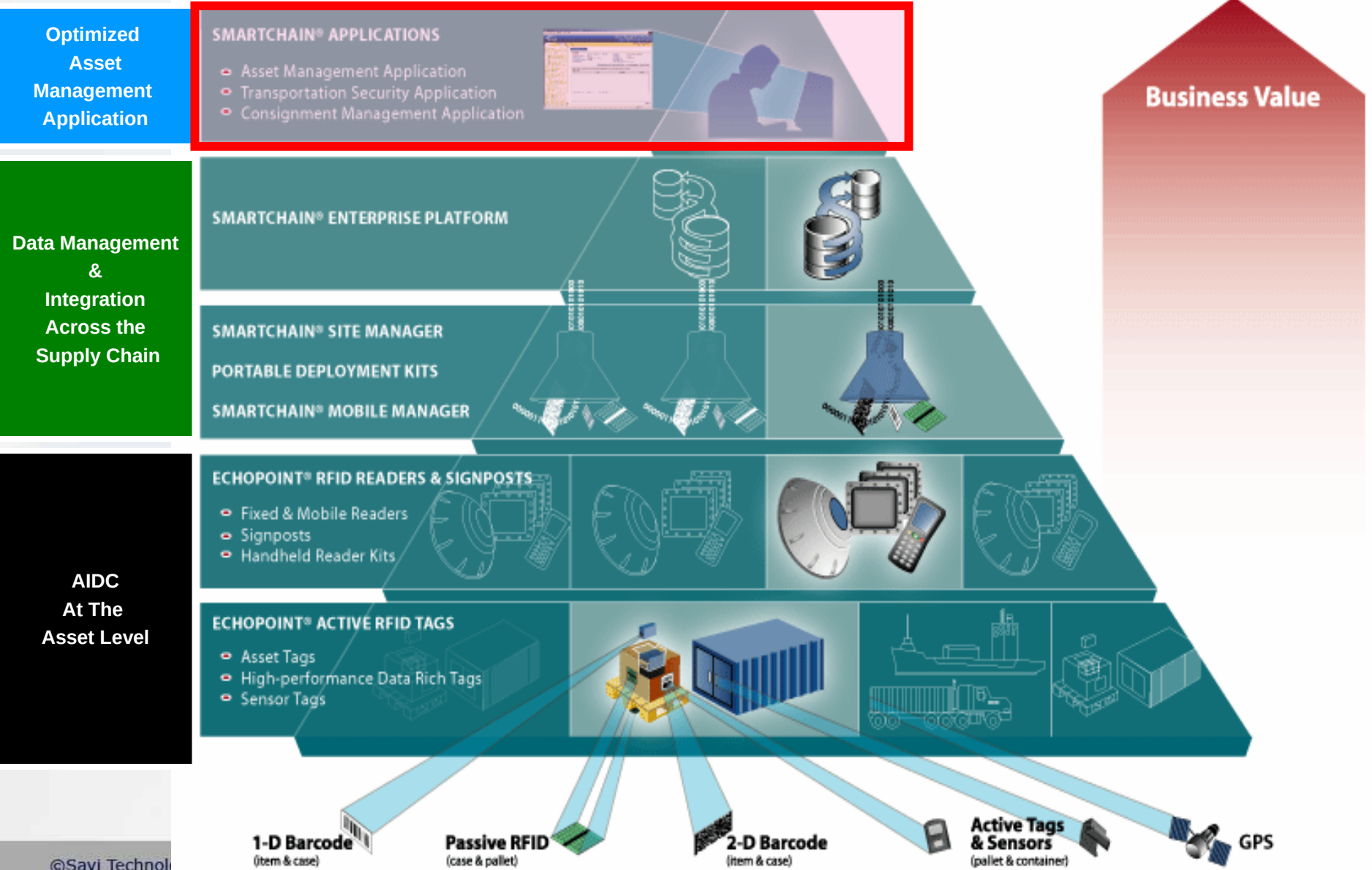
Optimized Asset Management Application

**Data Management & Integration
Across the Supply Chain**

**Automated Identification &
Data Capture Technology
At The Asset Level**



Savi Solutions Driving Value



Summary

- **Management of Transport Assets and their contents in the Retail Industry can be costly and problematic**
- **Retail Companies And Their Supply Chain Partners Are Increasingly Required to Introduce Traceability Programs**
- **Most Existing Systems Are Unable To Manage Assets & Provide Traceability In The Global Supply Chain**
- **Savi's Asset Management Solution Addresses These Industry Challenges**
- **Key Benefits Include:**
 - **Improved Asset Utilization**
 - Reducing Capital Investment, Lease, Rental & Demurrage Costs
 - **Improved Operational Efficiencies**
 - Lowering Operational Costs and Improved Customer Service
 - **Cost Effective Implementation of Standards Initiatives for Traceability**
 - Increasing Traceability For Sensitive Foodstuffs
 - Reducing Liability Risk & Increasing Brand Protection



Thank You



Contact:
ekgoh@ap.savi.com

